PRACTICAL 10

```
#include <stdio.h>
#include <stdlib.h>
#include <limits.h>

int main()
{
    int requests[] = {98, 183, 41, 122, 14, 124, 65, 67};
    int n = sizeof(requests) / sizeof(requests[0]);
    int head = 53;
    int visited[8] = {0}; // 0 = unvisited, 1 = visited
    int total_movement = 0;
    int current_pos = head;

    printf("SSTF Disk Scheduling - Step by Step Process\n");
    printf("-------------------------------------------\n");
    printf("Initial Head Position: %d\n\n", head);

    for (int i = 0; i < n; i++)
    {
        int min_distance = INT_MAX;
        int index = -1;

        // Find the closest unvisited request
        for (int j = 0; j < n; j++)
        {
            if (!visited[j])
            {
                int dist = abs(requests[j] - current_pos);
                if (dist < min_distance)
                {
                    min_distance = dist;
                    index = j;
                }
            }
        }

        // Mark as visited
        visited[index] = 1;
        total_movement += min_distance;

        printf("Step %d: Head at %3d -> Moving to %3d (Distance: %d)\n",
               i + 1, current_pos, requests[index], min_distance);

        current_pos = requests[index];
    }
```

```
    printf("\n-----------------------------------------\n");
    printf("Total Head Movement: %d cylinders\n", total_movement);

    return 0;
}
```

OUTPUT

Output

```
SSTF Disk Scheduling - Step by Step Process
-----------------------------------------
Initial Head Position: 53

Step 1: Head at  53 -> Moving to  41 (Distance: 12)
Step 2: Head at  41 -> Moving to  65 (Distance: 24)
Step 3: Head at  65 -> Moving to  67 (Distance: 2)
Step 4: Head at  67 -> Moving to  98 (Distance: 31)
Step 5: Head at  98 -> Moving to 122 (Distance: 24)
Step 6: Head at 122 -> Moving to 124 (Distance: 2)
Step 7: Head at 124 -> Moving to 183 (Distance: 59)
Step 8: Head at 183 -> Moving to  14 (Distance: 169)

-----------------------------------------
Total Head Movement: 323 cylinders


=== Code Execution Successful ===
```